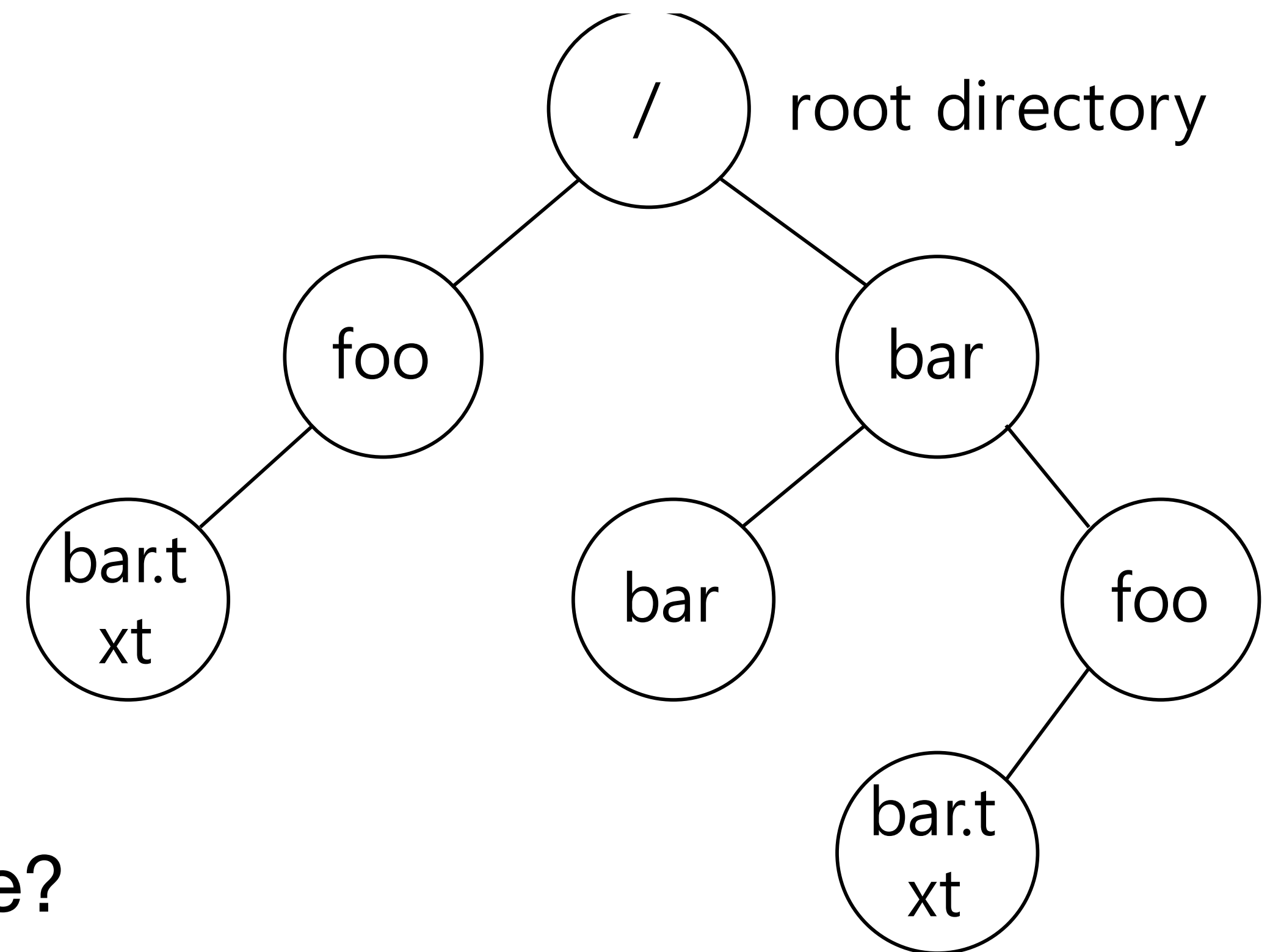


Review 5 (Optional)

Professor Amanda Bienz

Files and Directories

- Stored in a tree
- Step through directories to get to files
- Are files always leaves of the tree?
- Can directories ever be leaves of the tree?



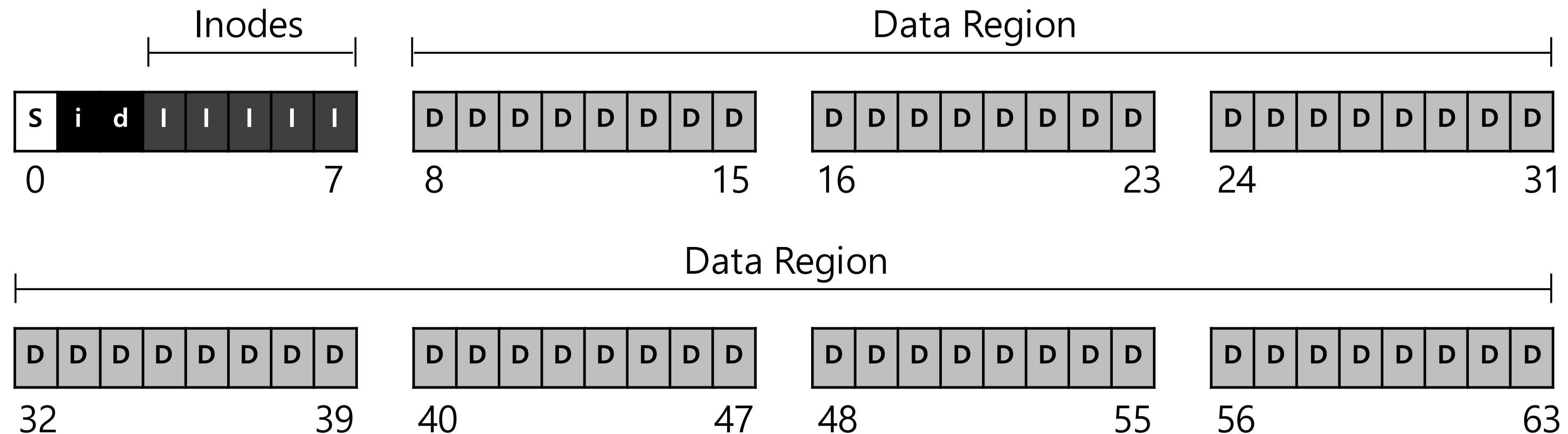
An Example Directory Tree

Some I/O Questions

- Is data instantly written to disk?
- What does `fsync()` do?
- Can two processes rename a file at the same time?

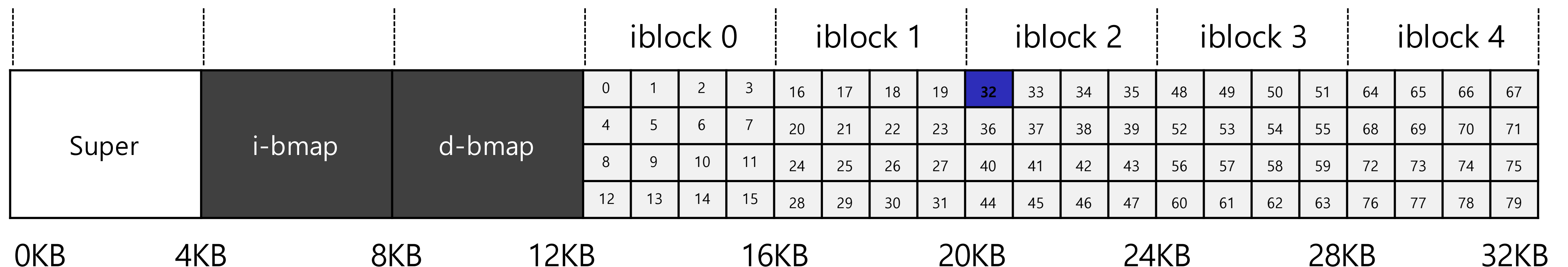
File System

- What is the superblock?
- What are inodes?
- What do inode and data bitmaps show?



An inode

- How do I find the location of a specific inode
- E.g. inode 32, inode size 256bytes
- How do I find the iblock number? The sector?
- E.g. if block size is 4KB, sector size is 512 bytes



Reading from disk

	data bitmap	inode bitmap	root inode	foo inode	bar inode	root data	foo data	bar data[0]	bar data[1]	bar data[2]
open(bar)			read	read	read	read	read			
read()					read			read		
read()					read				read	
read()					read					read

File Read Timeline (Time Increasing Downward)

Writing to Disk

	data bitmap	inode bitmap	root inode	foo inode	bar inode	root data	foo data	bar data[0]	bar data[1]	bar data[2]
create (/foo/bar)		read write	read	read	read write	read	read write			
write()	read write				read write			write		
write()	read write				read write				write	
write()	read write				read write					write

File Creation Timeline (Time Increasing Downward)

Buffering

- Why does the file system buffer writes?
- How can the file system improve performance of buffered writes?

Journaling

- What are the steps of journaling?
- When can crashes happen?
- How do you recover from these crashes?

Journaling

- What are the steps of journaling? **Write to journal, commit journal, checkpoint**
- When can crashes happen?
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Journaling

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- How do you recover from these crashes?

Journaling

- What are the steps of journaling? **Write to journal, commit journal, checkpoint**
- When can crashes happen? **While writing to journal, while checkpointing**
- How do you recover from these crashes? **Skip update, redo checkpoint**

Journaling

- Does data journaling help guarantee data is written to disk?
- What does journaling help guarantee?
- Is it necessary if you are not overwriting a block?

Snapshots

- Never overwrite blocks ... write new block and update pointers
- Keep old pointers (pointing to old blocks of data) to create a snapshot